Biliary Atresia, Part Three: Diagnosis. Gamma-Glutamyl Transferase, Matrix Metalloproteinase Seven, Ultrasound, Liver Biopsy and the Intraoperative Cholangiogram.

Picture a six-week-old boy referred on a Friday afternoon. His mother noticed that his stools have become pale, almost the colour of putty, and the referring doctor measured a conjugated bilirubin of three milligrams per decilitre. The questions arrive all at once. Is this biliary atresia? Which test should come first? Can we wait until Monday for the ultrasound? Should we send a serum marker and wait for the result, book a liver biopsy, or simply take him to theatre for a cholangiogram? Each of these choices costs time, and in biliary atresia time is liver. The older the infant at the Kasai operation, the lower the chance that bile will drain and that the child will keep his own liver.

The clinical question of this chapter is therefore simple to state. Which test ends the debate, and when? The main idea is that no test performed before surgery proves biliary atresia. Every blood test, scan and even the liver biopsy only estimates the probability of the disease. The only test that proves it is the intraoperative cholangiogram, an X-ray of the bile ducts taken during surgery, together with the microscopic examination of the bile duct remnant. The modern task is to move infants with a high probability to the operating room quickly, and to keep infants with a low probability away from an unnecessary operation. The tests should be run side by side, not one after another.

A note on sources. The framework comes from the 2017 joint guideline on cholestatic jaundice in infants by Rima Fawaz and colleagues, written for the North American Society for Pediatric Gastroenterology, Hepatology and Nutrition, known as NASPGHAN, and the European Society for Paediatric Gastroenterology, Hepatology and Nutrition, known as ESPGHAN. A newer comparison point is the 2025 Chinese guideline on the diagnosis and management of biliary atresia by Ge and colleagues, published in Hepatobiliary Surgery and Nutrition. The biomarker evidence comes from the 2017 discovery study by Lertudomphonwanit and colleagues in Cincinnati, a 2018 Chinese validation study by Yang and colleagues in Hepatology, a 2019 Taiwanese study by Wu and colleagues in the Journal of Pediatrics, and, most importantly, the 2024 North American validation by Pandurangi and colleagues from the Childhood Liver Disease Research Network, known as ChiLDReN, also published in Hepatology. Two meta-analyses, by He and colleagues in 2021 and by Wang and colleagues in 2025, pool the data. The imaging evidence comes from a 2016 meta-analysis of ultrasound by Zhou and colleagues, a 2009 Doppler study by Lee and colleagues in Radiology, and a 2022 meta-analysis of elastography by Wagner and colleagues. Neither the American Association for the Study of Liver Diseases, known as AASLD, nor the European Association for the Study of the Liver, known as EASL, has a

guideline on how to diagnose biliary atresia, so the paediatric documents carry the field.

Part one. Each test reads a different step.

It helps to see the disease as a chain of events, because each test reads a different link in that chain. The first link is that bile stops flowing out of the liver. Conjugated bilirubin, the water-soluble form of bilirubin that the liver has already processed, backs up into the blood, and the stools lose their colour because bile pigment no longer reaches the gut. We read this link with a fractionated bilirubin and by looking at the stools, ideally against a stool colour card. The second link is injury to the cells that line the bile ducts, the cholangiocytes. Injured cholangiocytes release two enzymes into the blood that we can measure: gamma-glutamyl transferase, usually shortened to GGT, and matrix metalloproteinase seven, usually shortened to MMP-7. The third link is a change in anatomy that an ultrasound can see: a small or irregular gallbladder, a fibrous cone of tissue at the hilum of the liver, and enlarged arteries. The fourth link is the liver's response to obstruction, which a pathologist sees in a biopsy as proliferating small bile ductules, plugs of bile, swelling of the portal tracts and fibrosis. The fifth link is the closed lumen itself, which only a cholangiogram demonstrates.

The tests at the start of this chain are sensitive but cannot name the cause, because any cholestatic liver disease raises conjugated bilirubin. The tests at the end of the chain are specific but invasive. Blood biomarkers and ultrasound sit in the middle, and their job is to decide who goes on to the invasive tests. One more principle matters. Tissue findings mature with age. A biopsy at three weeks of age may show only a picture of hepatitis, and MMP-7 is lower in the first month of life. A negative test early in life is therefore a reason to repeat or escalate, not a reason to stop.

Part two. Gamma-glutamyl transferase.

Gamma-glutamyl transferase is an enzyme on the surface of cholangiocytes. When the bile ducts are injured, its level rises. In most infants with biliary atresia it is clearly high, but it is also high in many other causes of infant cholestasis, so it narrows the field without deciding the question.

Three sets of numbers show how limited it is. In the 2018 Chinese study by Yang, a cut-off of 314 units per litre gave a sensitivity of only 64 percent and a specificity of 72 percent, with an area under the receiver operating characteristic curve of 0.72. The area under the curve is a single number that summarises how well a test separates two groups, where 0.5 is a coin toss and 1.0 is perfect separation. In the same infants, MMP-7 reached 0.99. In the large North American ChiLDReN cohort of 2024, a GGT cut-off of 267.5 units per litre gave a sensitivity of 86.6 percent and a specificity of 77.4 percent, with an area under the curve of 0.81. And in the 2021 meta-analysis by He, the pooled sensitivity of GGT was 80 percent and the pooled specificity 79 percent.

Why does GGT overlap so much? Because it signals cholangiocyte injury, not obstruction. Alagille syndrome, alpha-1 antitrypsin deficiency, progressive familial intrahepatic cholestasis type three, which is caused by variants in the ABCB4 gene, and neonatal sclerosing cholangitis all raise GGT. In neonatal sclerosing cholangitis the joint guideline notes values often above 800 international units per litre. GGT is also physiologically higher in newborns than in older children, so a value has to be read against age.

GGT is most useful in reverse. A normal or low GGT in a cholestatic infant points away from atresia and toward the low-GGT disorders: progressive familial intrahepatic cholestasis types one and two, defects of bile acid synthesis, and deficiency of tight junction protein two. Atresia with a low GGT does exist, but it is uncommon.

The threshold also climbs with age. A 2020 study by Shen and colleagues at Beijing Children's Hospital looked at 282 jaundiced infants under 120 days of age. In infants aged 60 days or younger, the best cut-off was about 191 units per litre, with a sensitivity of 89 percent and a specificity of 92 percent. Between 61 and 90 days the cut-off rose to about 213 units per litre, with a sensitivity of 85 percent and a specificity of 81 percent. Between 91 and 120 days it rose again to about 282 units per litre, with perfect sensitivity and a specificity of 93 percent, although that oldest group contained only seven infants with atresia, so the estimate is imprecise. When the authors called a test positive if either GGT or liver stiffness was above its cut-off, sensitivity rose to 96.3 percent but specificity fell to 58.8 percent, which is a screening rule, not a diagnosis. The 2025 Chinese guideline uses a simpler version of the same logic: consider atresia when GGT is above 100 units per litre, and regard a value above 300 units per litre as high suspicion.

Part three. Matrix metalloproteinase seven.

Matrix metalloproteinase seven is a protein-cutting enzyme made by injured cholangiocytes. It was found in 2017 by the Cincinnati group through a large screen of serum proteins, and in the mouse model it promotes bile duct injury rather than simply marking it. Because it reads epithelial injury, it rises early and stays high in atresia.

The 2018 Chinese validation study by Yang showed how wide the gap can be. The median level was about 2.9 nanograms per millilitre in healthy infants, about 11.5 in infants with other causes of cholestasis, and about 121 in infants with biliary atresia, roughly ten times higher than in the other cholestatic infants. At a cut-off of 52.85 nanograms per millilitre, sensitivity was 98.7 percent, which means 74 of the 75 infants with atresia were detected, and specificity was 95 percent. The area under the curve was 0.99.

Now comes the most important practical lesson of this chapter. The cut-off belongs to the assay, not to the disease. In the 2024 ChiLDReN study, the single-plex immunoassay used a cut-off of 52.8 nanograms per millilitre, but a second platform, a time-resolved fluorescence

resonance energy transfer assay, needed a cut-off of only 18.2 nanograms per millilitre for the same infants. In the 2019 Taiwanese study by Wu, which used a different commercial kit, the best cut-off was 1.43 nanograms per millilitre. A 2023 American study by Aleiri and colleagues, which used samples stored for a median of twelve years, needed a cut-off of 69 nanograms per millilitre and reached a sensitivity of only 68 percent. All of these numbers are correct for their own assays, and none can be moved onto another kit. The practical rule is to use only the threshold that has been validated for your own laboratory's platform, and to ask the laboratory which assay it runs.

How good is MMP-7 in a real-world Western population? The 2024 ChiLDReN study by Pandurangi enrolled 399 cholestatic infants from the network's prospective database, 201 with atresia and 198 without. MMP-7 had an area under the curve of 0.90, better than every other single test in the same infants. Liver biopsy read by pathologists reached 0.84, GGT reached 0.81, and stool colour only 0.68. At the 52.8 cut-off, sensitivity was 94 percent and specificity 77.8 percent. The positive predictive value, the chance that a positive test means atresia, was 64.5 percent, and the negative predictive value, the chance that a negative test means no atresia, was 96.8 percent. Six percent of infants with atresia fell below the cut-off, and 22 percent of infants without atresia were above it. The false positives are instructive: seven of eight infants with a choledochal cyst, eight of seventeen with alpha-1 antitrypsin deficiency, and thirteen of ninety-eight with indeterminate cholestasis had levels above the threshold. Specificity in this cohort was lower than in the Asian studies, probably because the controls were older and more varied and reflected the real mix of intrahepatic disease seen in North American centres.

The meta-analyses tell the same story. The 2021 meta-analysis by He, which pooled four studies of MMP-7, found a sensitivity of 96 percent and a specificity of 91 percent, with a diagnostic odds ratio of 313, compared with 15 for GGT. The 2025 meta-analysis by Wang, which pooled seventeen studies and 2,836 serum samples, found a sensitivity of 93 percent and a specificity of 85 percent, with an area under the curve of 0.96. Heterogeneity was high, with an I-squared of 78.6 percent, and it came mainly from studies published from 2023 onward. The pattern is clear: sensitivity stays in the low to mid nineties everywhere, while specificity falls as studies become larger, newer and more Western. The negative likelihood ratio in the Wang analysis was 0.08. In plain terms, if your pre-test probability of atresia is thirty percent, a clearly negative MMP-7 lowers it to about three percent.

MMP-7 has three weak points. First, levels are lower in the youngest infants. In the Wu study, the fourteen infants with atresia diagnosed at 30 days or younger had significantly lower levels than the twenty-two diagnosed later, and some overlapped with neonatal hepatitis. Second, some other conditions raise it above the cut-off, most importantly choledochal cyst and alpha-1 antitrypsin deficiency. Third, a small proportion of infants with atresia, about six percent in ChiLDReN, fall below the cut-off, and stored or

haemolysed samples may lower the result further. A low MMP-7 should never override pale stools. Finally, MMP-7 is usually a send-out test with a turnaround of days, so it must run in parallel with ultrasound, never in series.

So MMP-7 is best used to take atresia off the table when it is clearly low after the first month of life, which can spare an infant a biopsy or a laparotomy. A high value raises the probability, but it must be paired with anatomy and with an alpha-1 antitrypsin phenotype before anyone books a Kasai. The Wu study also found higher MMP-7 six months after the Kasai in children who later needed a transplant, but this is not yet a validated monitoring test.

Part four. Ultrasound and other imaging.

A fasting abdominal ultrasound is the first imaging test in every cholestatic infant. It excludes a choledochal cyst and gallstones and can show signs of atresia. The 2016 meta-analysis by Zhou, which included twenty-three studies, gives the numbers. The triangular cord sign, a triangular or tubular echogenic cone of fibrous tissue just above the bifurcation of the portal vein, had a pooled sensitivity of 74 percent and a specificity of 97 percent. In other words, when you see it, atresia is very likely, but it is missed in about a quarter of affected infants. Gallbladder abnormalities had a sensitivity of 85 percent and a specificity of 92 percent. Combining the two signs raised sensitivity to 95 percent at the cost of a lower specificity of 89 percent.

Doppler adds a further sign. In 2009 Lee and colleagues in Korea described hepatic subcapsular flow, small arterial vessels running just beneath the liver capsule, seen with colour Doppler. It was present in all twenty-nine infants with atresia, with a specificity of 86 percent. A later meta-analysis of nine studies pooled a sensitivity of 95 percent and a specificity of 92 percent. An enlarged hepatic artery is another supportive finding.

The gallbladder deserves careful attention. It may be absent, small or irregular. The so-called gallbladder ghost triad, described by Tan Kendrick and colleagues in 2003, combines a short gallbladder, a thin or absent mucosal lining, and an irregular, lobulated contour. A gallbladder that does not contract after a feed is also suspicious. Two traps matter here. The infant must be fasted for three to four hours, otherwise every gallbladder looks small. And a normal-sized, contracting gallbladder lowers the probability of atresia but does not abolish it. The joint guideline is explicit that a normal ultrasound does not exclude atresia.

Two other ultrasound findings change surgical planning. A cyst at the porta hepatis may be cystic atresia or a choledochal cyst; non-dilated intrahepatic ducts favour atresia, while dilated ducts favour a cyst. And signs of the splenic malformation syndrome, such as polysplenia, situs anomalies, a preduodenal portal vein or an absent inferior vena cava,

should be reported to the surgeon before the operation. Repeating a normal ultrasound a week later rarely changes management; it is better to escalate to biomarkers, biopsy or cholangiography.

Shear wave elastography, which measures liver stiffness with ultrasound, is an adjunct. The 2022 meta-analysis by Wagner pooled eleven studies and 1,307 infants and found a sensitivity of 83 percent and a specificity of 79 percent, with an area under the curve of 0.91. Its additional value is that stiffness also estimates fibrosis before the Kasai.

Hepatobiliary scintigraphy, a nuclear medicine scan that follows a tracer from the liver into the bowel, is very sensitive but not specific. In the meta-analysis cited by the joint guideline, a non-draining scan had a sensitivity of 98.7 percent and a specificity of only 70.4 percent. Excretion of tracer into the bowel argues against atresia, but failure to excrete happens in many severe intrahepatic diseases. Many centres prime the infant with phenobarbital for five days before the scan to increase excretion, and the guideline notes that this often delays diagnosis and the Kasai without proven benefit. Magnetic resonance cholangiopancreatography, known as MRCP, has a reported sensitivity of 99 percent but a specificity of only 36 percent, because the tiny ducts of an infant are often not seen. Endoscopic retrograde cholangiopancreatography, known as ERCP, can reach a sensitivity between 86 and 100 percent and a specificity between 87 and 94 percent, but it needs an expert endoscopist, special infant equipment and general anaesthesia. Percutaneous cholecystocholangiography, in which contrast is injected into the gallbladder through the skin under ultrasound guidance, is reported in the Chinese guideline with a sensitivity of 100 percent and a specificity of 87 percent where expertise exists. The common thread is that tests with high sensitivity and low specificity are useful only when they are negative. A non-draining scan or a duct that is not seen still needs a definitive test.

Part five. Liver biopsy.

A percutaneous liver biopsy reads the liver's response to obstruction. The obstructive pattern consists of proliferation of small bile ductules at the edges of the portal tracts, plugs of bile inside those ductules, portal oedema and fibrosis, with preserved architecture of the liver lobule. The hepatitic pattern of idiopathic neonatal hepatitis looks different: lobular disarray, prominent multinucleated giant cells, portal inflammation and little ductular change. There is overlap. Giant cells are seen in twenty to fifty percent of infants with atresia, and alpha-1 antitrypsin deficiency, Alagille syndrome, progressive familial intrahepatic cholestasis type three, cystic fibrosis and parenteral nutrition can all produce ductular changes.

In experienced hands the biopsy is accurate. The joint guideline states that experienced paediatric pathologists give the correct diagnosis in ninety to ninety-five percent of cases, and it calls an appropriately timed biopsy showing ductular proliferation, bile plugs and fibrosis the most supportive test, with a grade 1B recommendation. In the research

consortium that preceded ChiLDReN, the proportion of proven atresia cases that individual pathologists called obstructive ranged from 79 to 98 percent, with a positive predictive value of 90.7 percent. In the 2024 ChiLDReN comparison, pathology reached an area under the curve of 0.84, below MMP-7 at 0.90. The complication rate is about 1.7 percent in large paediatric series and higher in small infants, so vitamin K must be given and coagulation checked first. The main caveat is timing: a biopsy taken too early may show only hepatitis. If the stools remain pale, the biopsy should be repeated or the infant should go to cholangiography.

Part six. The intraoperative cholangiogram.

The joint guideline gives a grade 1A recommendation that the intraoperative cholangiogram, together with histology of the duct remnant, is the gold standard, and the 2025 Chinese guideline agrees. The surgeon first inspects the liver and gallbladder. An atrophic gallbladder containing white bile or no bile, and a fibrous porta, make atresia likely. Contrast is then injected through the gallbladder. If it flows into the duodenum and fills the hepatic ducts, the infant does not have atresia. If the distal ducts fill but the proximal ducts do not, or if there is no lumen at all, the diagnosis is atresia, and the surgeon proceeds to the portoenterostomy under the same anaesthetic and sends the remnant to pathology. A bile-free cyst at the porta with abnormal intrahepatic ducts is cystic atresia, not a choledochal cyst.

The cholangiogram has its own traps. A small but open biliary tree in Alagille syndrome, cystic fibrosis or severe intrahepatic cholestasis may fail to fill with contrast and mimic atresia. For this reason the alpha-1 antitrypsin phenotype, a sweat test or genetic testing for cystic fibrosis, and a search for features of Alagille syndrome should be completed before surgery. When the probability of atresia is only moderate, a laparoscopic cholangiogram can avoid a full laparotomy. The cholangiogram should be planned and consented as the first step of a Kasai, so that one trip to theatre does both jobs.

Part seven. Putting it together, and what the guidelines say.

An integrated pathway takes days, not weeks. The first step is suspicion: a conjugated bilirubin above 1.0 milligram per decilitre at any age, pale stools on a stool colour card, and, on the first day, exclusion of the urgent treatable conditions of sepsis, galactosaemia, hypothyroidism and hypopituitarism. The second step is laboratory work: GGT, an alpha-1 antitrypsin level and phenotype, coagulation, and MMP-7 on a validated platform. The third step is a fasting ultrasound looking for the gallbladder, the triangular cord, subcapsular flow, a cyst and the splenic malformation. The second and third steps run in parallel. The fourth step is a decision. With a high probability, the infant goes to a cholangiogram. With an intermediate probability, an expert biopsy helps. With a clearly low MMP-7 and evidence of bile flow, the work-up turns to other causes. The fifth step is surgery: the cholangiogram and

the Kasai in one session, with genetic testing running alongside. Give vitamin K before any invasive step. And remember that ordering ultrasound, biomarkers and biopsy one after another, each waiting for the last, is how a forty-day-old infant becomes a seventy-day-old Kasai.

How do the guidelines compare? On the bilirubin trigger, the 2017 joint guideline and the 2025 Chinese guideline are aligned: a conjugated bilirubin above about one milligram per decilitre is abnormal, and the Chinese guideline adds that it should make up more than twenty percent of the total. On GGT they differ in emphasis. The joint guideline says only that it is often high in atresia and that a low value suggests other diseases, while the Chinese guideline gives numerical thresholds of 100 and 300 units per litre. On MMP-7 there is a material difference. The 2017 guideline predates the evidence and does not mention it, while the Chinese guideline calls it a potential early marker with a weak grade 2C recommendation and declines to endorse a cut-off because thresholds are inconsistent, and the 2024 ChiLDReN study now provides North American validation. On ultrasound they are aligned: first line, suggestive but not diagnostic. On scintigraphy they are aligned: not a stand-alone test. On elastography they differ, since only the Chinese guideline addresses it, calling it useful with a grade 2B recommendation. On biopsy they differ: the joint guideline calls it the most supportive test with grade 1B, while the Chinese guideline says only that it might help, with grade 2B. On the cholangiogram they are aligned: it is the definitive test. The material gap is that the main Western guideline predates the best-validated serum test, and an update should define assay-specific thresholds and the place of MMP-7 before biopsy.

Part eight. Traps and pearls.

Four reasoning errors cause most of the harm. The first is to say that a normal ultrasound excludes atresia; some infants with atresia have a visible gallbladder, the triangular cord is missed in a quarter, and the stool colour outranks the scan. The second is to book a Kasai because MMP-7 is high; choledochal cyst and alpha-1 antitrypsin deficiency also raise it, so check anatomy and phenotype and let the cholangiogram decide. The third is to close the case because a biopsy at three weeks showed neonatal hepatitis; early atresia can look hepatitic, and giant cells do not exclude it. The fourth is to prime with phenobarbital and wait five days for a scan whose specificity is about seventy percent, which buys delay without a decision. A fifth error is to apply a published MMP-7 cut-off to a different laboratory kit, which can turn a clear result into a false one in either direction.

Let me close with how this changes practice, through a few infants. A six-week-old with pale stools, a GGT of 410 units per litre, an MMP-7 of 150 nanograms per millilitre on a validated assay and an absent gallbladder needs vitamin K and a cholangiogram with a Kasai this week; a biopsy adds nothing. A five-week-old with a GGT of 90 units per litre, a

normal contracting gallbladder and an MMP-7 below the cut-off is unlikely to have atresia, and the work-up should turn to progressive familial intrahepatic cholestasis, bile acid synthesis defects and genetics. A three-week-old with a conjugated bilirubin of 2.1 milligrams per decilitre, a borderline MMP-7 and a hepatitic biopsy should not be discharged from suspicion; repeat the MMP-7 or the biopsy within one to two weeks and keep watching the stools. An eight-week-old with a high MMP-7 and a cyst at the porta with dilated intrahepatic ducts more likely has a choledochal cyst and needs MRCP or a cholangiogram, not a Kasai by default. An infant with a high MMP-7, a PiZZ alpha-1 antitrypsin phenotype and pigmented stools has alpha-1 antitrypsin deficiency and should not have a laparotomy. And an infant with polysplenia and situs inversus on the scan and a rising conjugated bilirubin probably has the splenic malformation form of atresia; go to cholangiography and warn the surgeon about the vascular anatomy.

So tomorrow in clinic, remember these points. No test before surgery proves atresia; the cholangiogram with remnant histology does. GGT separates atresia from other cholestasis only moderately, with areas under the curve between about 0.72 and 0.81, and its useful thresholds rise from about 190 to about 280 units per litre with age. MMP-7 in atresia is about ten times higher than in other cholestasis, and in the North American cohort it reached an area under the curve of 0.90, a sensitivity of 94 percent and a negative predictive value of 97 percent. Its cut-off depends on the assay, it is lower in infants under a month old, and it is also high in choledochal cyst and alpha-1 antitrypsin deficiency. On ultrasound, the triangular cord is specific but often missed, gallbladder abnormalities and subcapsular flow are more sensitive, and a normal scan does not exclude atresia. Scintigraphy is sensitive but not specific, and priming only delays. A well-timed biopsy read by an expert is right in about nine of ten cases. Above all, run the tests in parallel and plan the cholangiogram as the start of the Kasai, because every week saved is liver saved.